

Week 07 – Jacobians

ELEC0129 Introduction to Robotics

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Schedule

Legend:

A: MPEB 6.02 Teaching Lab

B: Roberts 7.04 PC Lab

Week	Recorded, uploaded by Friday of previous week	Workshop, Mondays 11am-1pm and/or Wednesdays 9am-11am	Notes
1	(Scenario Week)		
2	Lec: Intro; Spatial description	A: Workshop: Offline programming	Half class Mon, half class Wed
3	Lec & Tut: Spatial description	A: Workshop: Build robot	
4	Lec & Tut: Forward kinematics	A: Workshop: Forward kinematics	
5	Lec & Tut: Inverse kinematics	B: Workshop: Offline programming	Half class Mon, half class Wed
RW	(Reading Week)		
6	(Scenario Week)		
7	Lec & Tut: Jacobians	A: Workshop: Inverse kinematics	
8	Lec: Trajectory Planning	A: Workshop: Trajectory planning	
9	Lec & Tut: Dynamics	A: Workshop: Trajectory planning	
10	Lec & Tut: Control	A: Workshop: Pick-and-place demo	

Content

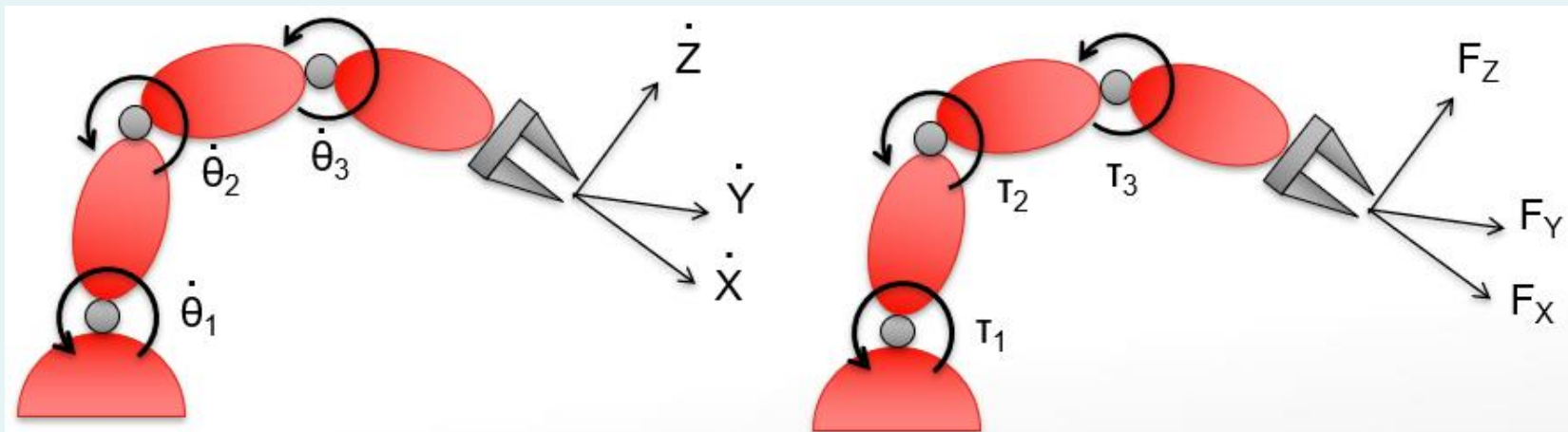
- Introduction
- Velocity Propagation
- Linear Jacobian
- Singularities
- Static Forces in Manipulators
- Tutorial Questions

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Introduction

- In this lecture, we will learn:
 - Relationship between **joint velocity** and in **Cartesian velocity**.
 - Relationship between the **joint torques** and the **static force** applied by end-effector onto a work surface.

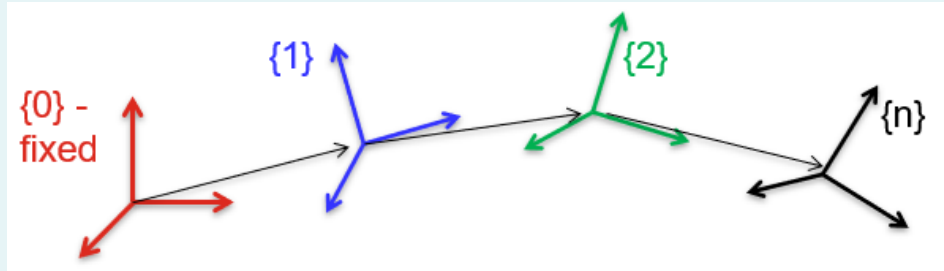


Content

- Introduction
- **Velocity Propagation**
- Linear Jacobian
- Singularities
- Static Forces in Manipulators
- Tutorial Questions

Velocity Propagation

- The robotic manipulator is a **chain of bodies**, each capable of moving relative to its neighbour.

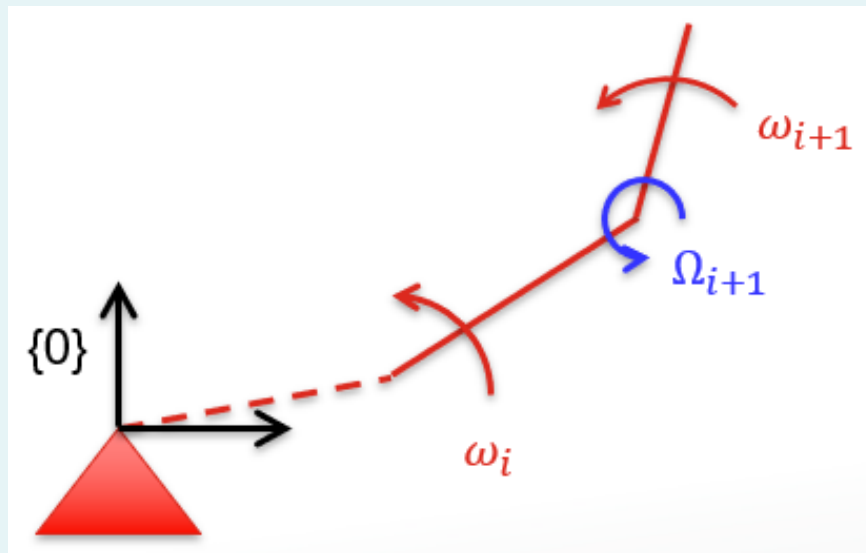


- Therefore, the velocity of each link can be computed in order, starting from the base.

Velocity of link $i+1$ = Velocity of link i + new components due to joint $i+1$

Notations – Rotational Velocities (1)

- We need to differentiate between **relative velocity** and **absolute velocity**.
- For **rotational velocity**:
 - Ω_{i+1} : Angular speed of joint $i+1$ wrt joint i , “**relative**”.
 - ω_i, ω_{i+1} : “**Absolute**” velocity wrt. frame $\{0\}$.



Notations – Rotational Velocities (2)

- How to get ω and Ω ?
- We knew: Relative rotation from frame $\{i\}$ to $\{i + 1\}$:

$${}^i_{i+1}R$$

- Relative angular velocity of $\{i + 1\}$ wrt $\{i\}$ is written as:

$${}^i\Omega_{i+1}$$

- This is a 3x1 vector.
- Direction = instantaneous axis of rotation
- Magnitude = speed

Notations – Rotational Velocities (3)

- We can change the reference frame of expression using the rotation matrix.
- E.g. Relative angular velocity of $\{i + 1\}$ wrt $\{i\}$, expressed in frame $\{i + 1\}$ is:

$${}^{i+1}\Omega_{i+1} = {}^{i+1}_i R \cdot {}^i\Omega_{i+1}$$

- If we now change the reference frame of expression to frame $\{0\}$, we then get the absolute velocity.

$${}^0\Omega_{i+1} = {}^0_i R \cdot {}^i\Omega_{i+1} \triangleq \omega_{i+1}$$

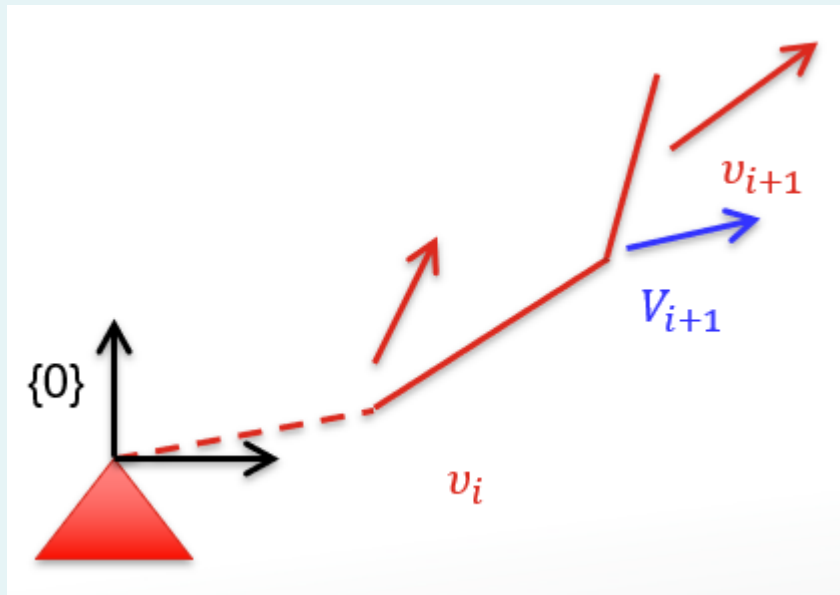
Notations – Rotational Velocities (4)

- Even the absolute velocity can have a change of reference frame, e.g.

$${}^{i+1}\omega_{i+1} = {}^{i+1}({}^0\Omega_{i+1}) = {}^{i+1}_0R \cdot {}^0\Omega_{i+1} = {}^{i+1}_0R \cdot \omega_{i+1}$$

Notations – Linear Velocities (1)

- For **linear** velocity:
 - V_{i+1} : Linear speed of joint $i+1$ wrt joint i , “**relative**”.
 - v_i, v_{i+1} : “**Absolute**” velocity wrt. frame $\{0\}$.



Notations – Linear Velocities (2)

- How to get v and V ?
- We knew: Relative position from frame $\{i\}$ to $\{i + 1\}$:

$${}^iP_{i+1}$$

- Relative linear velocity of $\{i + 1\}$ wrt $\{i\}$ is written as:

$$\frac{d}{dt} ({}^iP_{i+1}) = {}^iV_{i+1}$$

Notations – Linear Velocities (3)

- We can change the reference frame of expression using the rotation matrix.
- E.g. Relative linear velocity of $\{i + 1\}$ wrt $\{i\}$, expressed in frame $\{i + 1\}$ is:

$${}^{i+1}V_{i+1} = {}^{i+1}_i R \cdot {}^iV_{i+1}$$

- If we now change the reference frame of expression to frame $\{0\}$, we then get the absolute velocity.

$${}^0V_{i+1} = {}^0_i R \cdot {}^iV_{i+1} \triangleq v_{i+1}$$

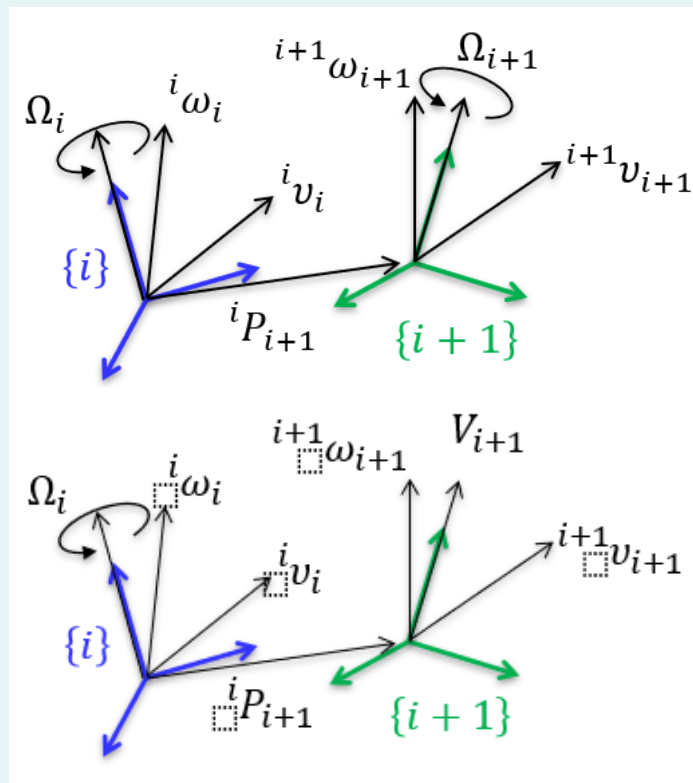
Notations – Linear Velocities (4)

- Even the absolute velocity can have a change of reference frame, e.g.

$${}^{i+1}v_{i+1} = {}^{i+1}({}^0V_{i+1}) = {}^{i+1}_0R \cdot {}^0V_{i+1} = {}^{i+1}_0R \cdot v_{i+1}$$

Vel. Propagation Algorithm: Derivation

- Consider links i and $i + 1$, together with their velocities.
- We want to find out the **velocities of current frame ($i + 1$)**, as the result of:
 - Velocities of **previous frame (i)**;
 - Velocities provided by the **joint $i + 1$** .

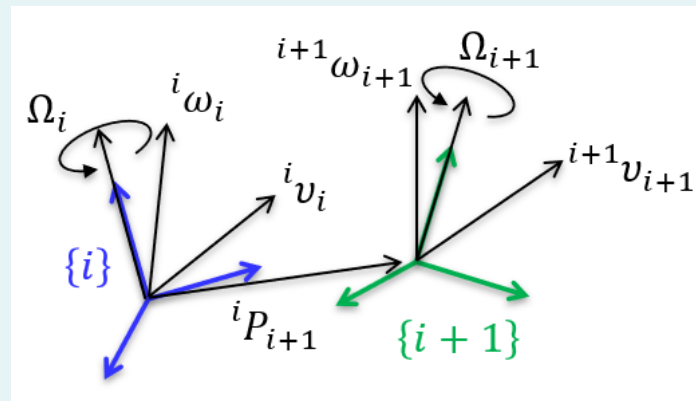


Rotational Velocity Propagation (1)

- If joint $i + 1$ is revolute, rotational velocities can be added, if both ω vectors are expressed in the same frame. Therefore:

$$\begin{aligned} {}^i\omega_{i+1} &= {}^i\omega_i + {}_{i+1}^iR \cdot {}^{i+1}\Omega_{i+1} \\ &= {}^i\omega_i + {}_{i+1}^iR \cdot \dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1} \end{aligned}$$

- $\dot{\theta}_{i+1}$ is a scalar;
- ${}^{i+1}\hat{Z}_{i+1} = [0 \ 0 \ 1]^T$;
- Thus: $\dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1} = [0 \ 0 \ \dot{\theta}_{i+1}]^T$.



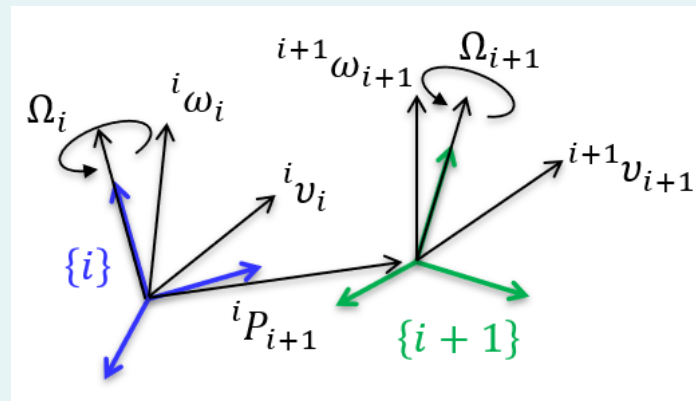
Rotational Velocity Propagation (2)

- Premultiply the first equation on the previous slide with ${}^{i+1}_i R$ gives:

*

$${}^{i+1}\omega_{i+1} = {}^{i+1}_i R \cdot {}^i\omega_i + \dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1}$$

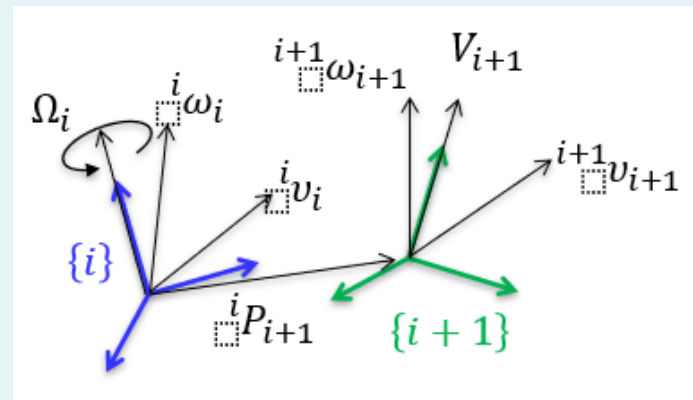
- The reason for doing this is to get the **superscript on the left hand side** as $i + 1$, which is useful for the iterative algorithm later.



Rotational Velocity Propagation (3)

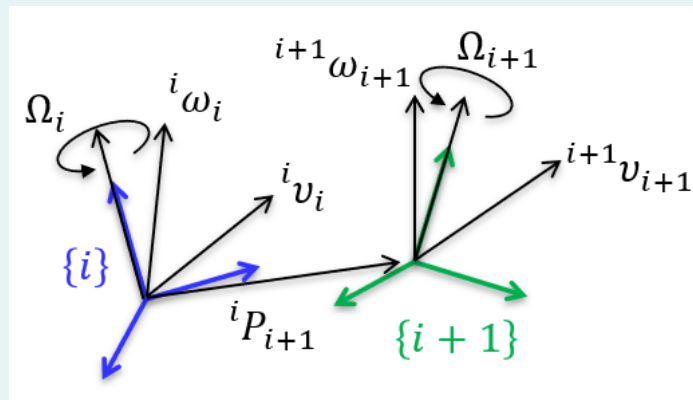
- If the joint $i + 1$ is prismatic, then there is no new rotational velocity contributed by the joint.

*
$${}^{i+1}\omega_{i+1} = {}^{i+1}_i R \cdot {}^i\omega_i$$

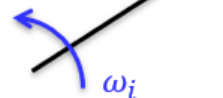


Linear Velocity Propagation (1)

- As for linear velocities, if the joint is **revolute**, then there is no linear velocity between the two frames.
- However, the rotation at joint i creates a **tangential linear velocity** at frame $i + 1$.



Tangential
linear velocity
caused by
rotation of
previous joint



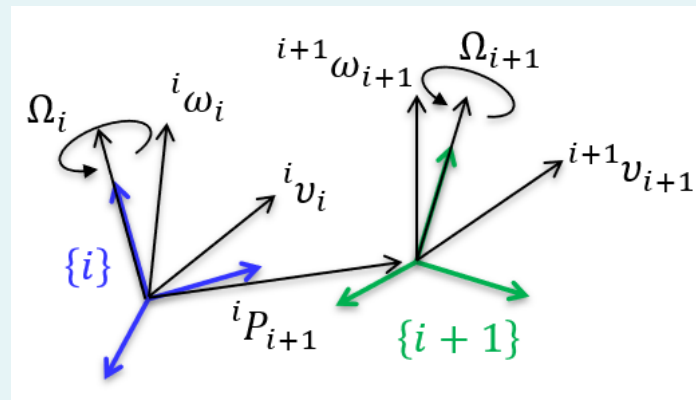
Linear Velocity Propagation (2)

- Therefore:
- The linear velocity of the origin of $\{i + 1\}$ = linear velocity of origin of $\{i\}$ + new component caused by rotational velocity of link i .

$${}^i v_{i+1} = {}^i v_i + {}^i \omega_i \times {}^i P_{i+1}$$

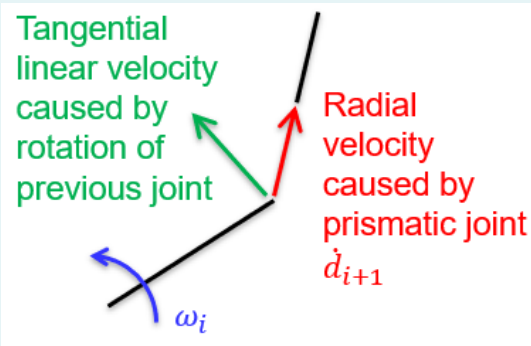
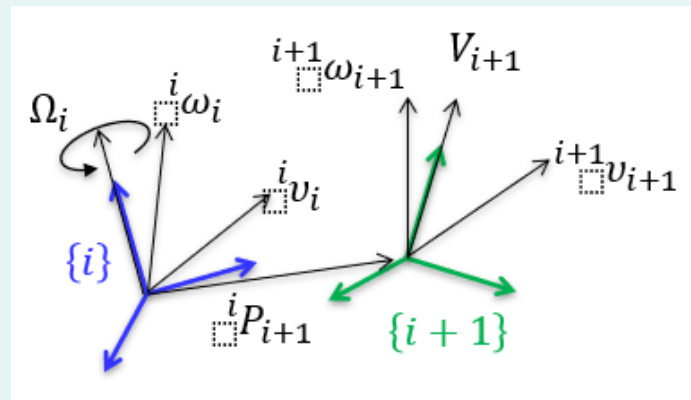
- Premultiply each term with ${}^{i+1}_i R$:

*
$${}^{i+1} v_{i+1} = {}^{i+1}_i R \cdot ({}^i v_i + {}^i \omega_i \times {}^i P_{i+1})$$



Linear Velocity Propagation (3)

- If the joint is **prismatic**, then the propagated linear velocity is the **sum of**:
 - The **tangential** linear velocity at joint $i + 1$ caused by rotation of previous joint;
 - The **radial** linear velocity created by the prismatic joint $i + 1$.



*

$${}^{i+1}v_{i+1} = {}^{i+1}_i R \cdot ({}^i v_i + {}^i \omega_i \times {}^i P_{i+1}) + \dot{d}_{i+1} \cdot {}^{i+1} \hat{Z}_{i+1}$$

Vel Propagation Algorithm Summary (1)

- Start with ${}^0\omega_0, {}^0v_0$.
- Calculate recursively from link to link:
 - For revolute joints:

$$\begin{aligned} {}^{i+1}\omega_{i+1} &= {}^{i+1}_iR \cdot {}^i\omega_i + \dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1} \\ {}^{i+1}v_{i+1} &= {}^{i+1}_iR \cdot ({}^iv_i + {}^i\omega_i \times {}^iP_{i+1}) \end{aligned}$$

- For prismatic joints:

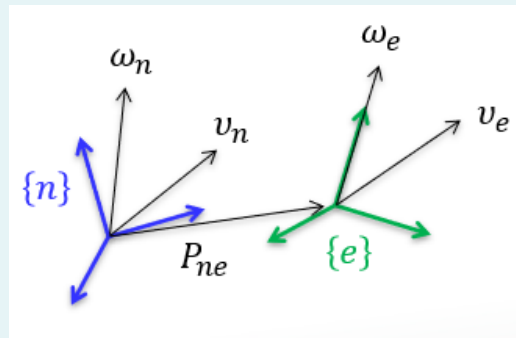
$$\begin{aligned} {}^{i+1}\omega_{i+1} &= {}^{i+1}_iR \cdot {}^i\omega_i \\ {}^{i+1}v_{i+1} &= {}^{i+1}_iR \cdot ({}^iv_i + {}^i\omega_i \times {}^iP_{i+1}) + \dot{d}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1} \end{aligned}$$

- Continue until ${}^n\omega_n, {}^nv_n$.

Vel Propagation Algorithm Summary (2)

- The calculation so far ends at the last frame.
- If the **end-effector** is of interest, we can propagate the velocity from the last link to the end-effector as well:
 - The end-effector is usually fixed onto the last link, and there is no motion relative to them.
 - The velocity of end-effector is thus simply:

$$\begin{aligned} {}^n\omega_e &= {}^n\omega_n \\ {}^nv_e &= {}^nv_n + {}^n\omega_n \times {}^nP_e \end{aligned}$$



Vel Propagation Algorithm Summary (3)

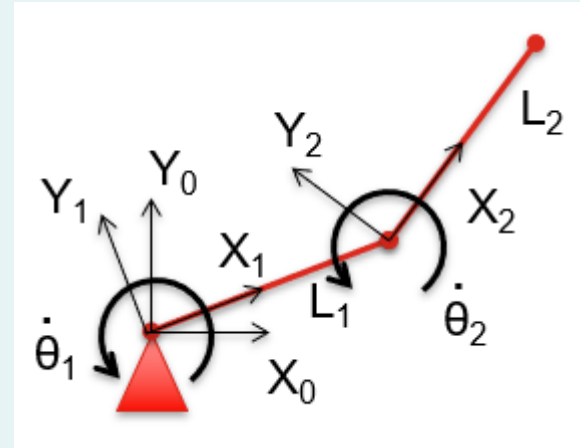
- Finally, transform ${}^n\omega_n$, ${}^n\mathbf{v}_n$ or ${}^n\omega_e$, ${}^n\mathbf{v}_e$ back into the base frame by:

$$\begin{aligned} {}^0\mathbf{v}_n &= {}^0_nR \cdot {}^n\mathbf{v}_n \\ {}^0\omega_n &= {}^0_nR \cdot {}^n\omega_n \end{aligned}$$

$$\begin{aligned} {}^0\mathbf{v}_e &= {}^0_nR \cdot {}^n\mathbf{v}_e \\ {}^0\omega_e &= {}^0_nR \cdot {}^n\omega_e \end{aligned}$$

Example (1)

- Given a two link robot as shown.
- What is the velocity at its tip?
- Because the formulae need the transformation matrices, they are computed first.



$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & L_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

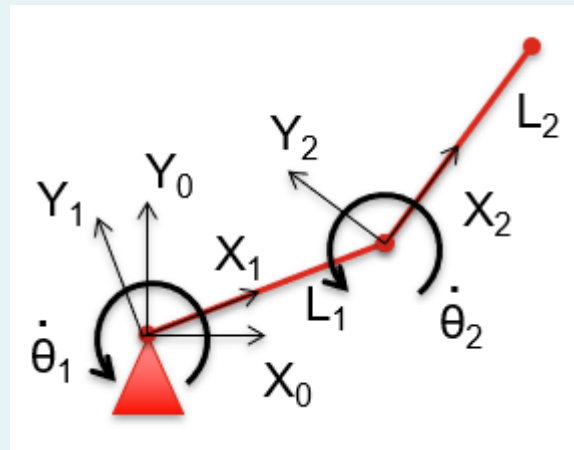
Example (2)

- Next, we apply the iterative formula.
- Start with:

$${}^0\omega_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad {}^0v_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- Then:

$${}^1\omega_1 = {}^1_0R \cdot \underbrace{{}^0\omega_0}_0 + \dot{\theta}_1 \cdot {}^1\hat{Z}_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

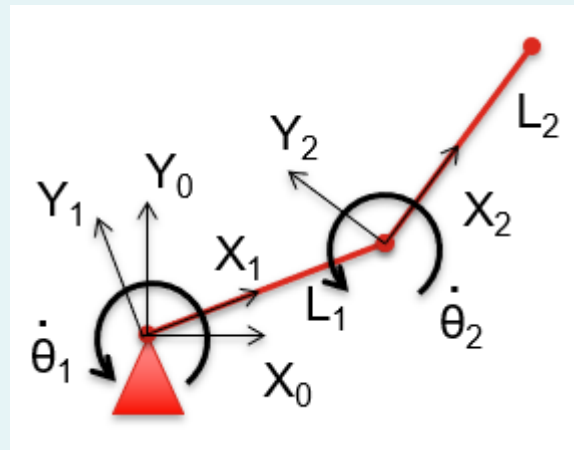


Example (3)

- Also:

$$\begin{aligned} {}^1v_1 &= {}^1_0R \cdot \left(\underbrace{{}^0v_0}_0 + \underbrace{{}^0\omega_0}_0 \times {}^0P_1 \right) \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

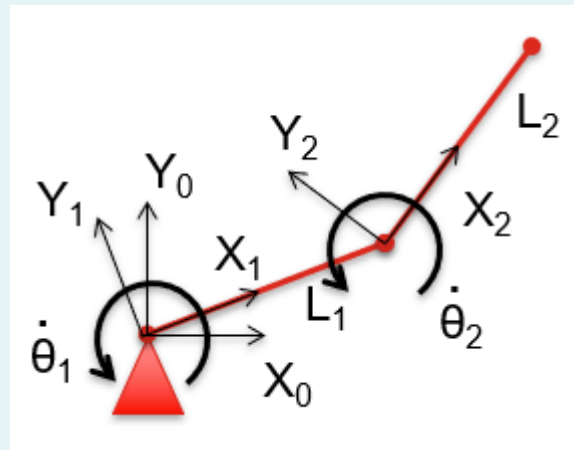
- Makes sense! Frame {1} doesn't have linear velocity.



Example (4)

- We move on to frame {2}:

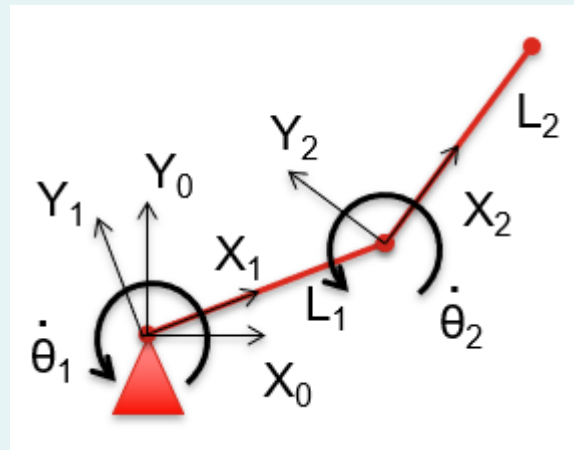
$$\begin{aligned} {}^2\omega_2 &= {}^2_1R \cdot {}^1\omega_1 + \dot{\theta}_2 \cdot {}^2\hat{Z}_2 \\ &= {}^2_1R^T \cdot {}^1\omega_1 + \dot{\theta}_2 \cdot {}^2\hat{Z}_2 \\ &= \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \end{aligned}$$



Example (5)

• Also:

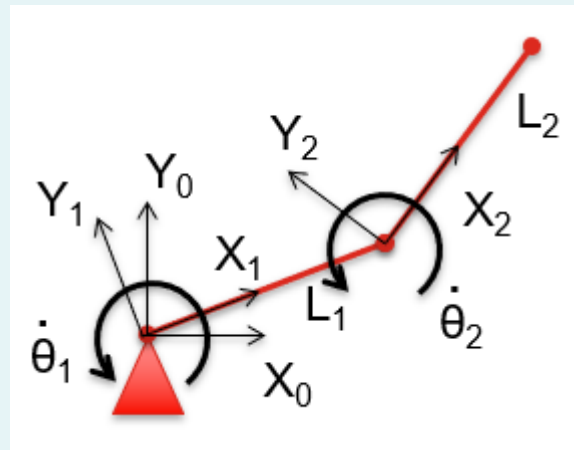
$$\begin{aligned}
 {}^2v_2 &= {}^1R \cdot ({}^1v_1 + {}^1\omega_1 \times {}^1P_2) \\
 &= {}^1R^T \cdot ({}^1v_1 + {}^1\omega_1 \times {}^1P_2) \\
 &= \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{vmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ L_1 & 0 & 0 \end{vmatrix} \right) \\
 &= \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ L_1 \dot{\theta}_1 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} L_1 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 \\ 0 \end{bmatrix}
 \end{aligned}$$



Example (6)

- Finally, we can calculate the velocities at the tip of robot:

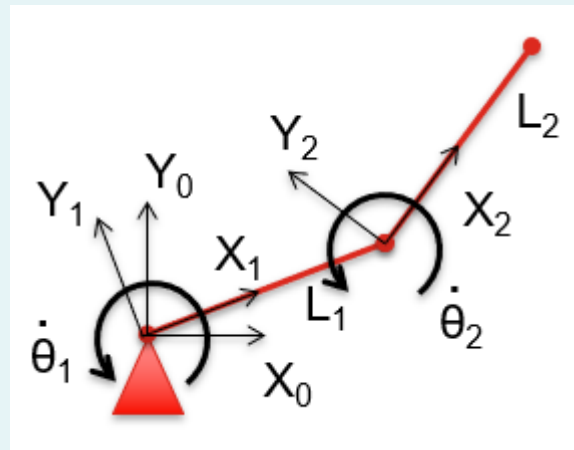
$$\begin{aligned}
 {}^2\omega_e &= {}^2\omega_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \\
 {}^2v_e &= {}^2v_2 + {}^2\omega_2 \times {}^2P_e \\
 &= \begin{bmatrix} L_1 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 \\ 0 \end{bmatrix} + \begin{vmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 + \dot{\theta}_2 \\ L_2 & 0 & 0 \end{vmatrix} \\
 &= \begin{bmatrix} L_1 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 + L_2 (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}
 \end{aligned}$$



Example (7)

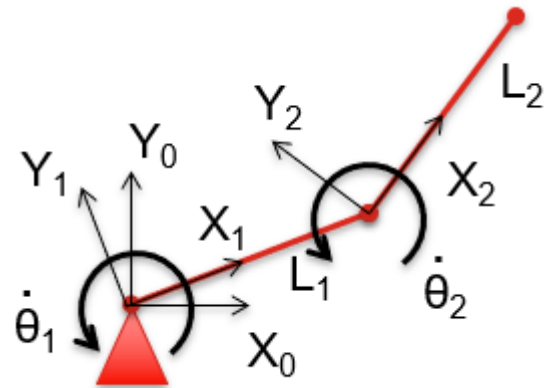
- Transform the results back to base frame:

$$\begin{aligned} {}^0\omega_e &= {}^0_2R \cdot {}^2\omega_e \\ &= {}^0_1R \cdot {}^1_2R \cdot {}^2\omega_e \\ &= \begin{bmatrix} c_{12} & -s_{12} & 0 \\ s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \end{aligned}$$



Example (8)

$$\begin{aligned}
 {}^0v_e &= {}^0_2R \cdot {}^2v_e \\
 &= {}^0_1R \cdot {}^1_2R \cdot {}^2v_e \\
 &= \begin{bmatrix} c_{12} & -s_{12} & 0 \\ s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} L_1 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 + L_2 (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} L_1 c_{12} s_2 \dot{\theta}_1 - L_1 s_{12} c_2 \dot{\theta}_1 - L_2 s_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ L_1 s_{12} s_2 \dot{\theta}_1 + L_1 c_{12} c_2 \dot{\theta}_1 + L_2 c_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} -L_1 s_1 \dot{\theta}_1 - L_2 s_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ L_1 c_1 \dot{\theta}_1 + L_2 c_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}
 \end{aligned}$$



Example (9)

- The results can be summarized in matrix forms:

$$\begin{bmatrix} ({}^0\omega_e)_x \\ ({}^0\omega_e)_y \\ ({}^0\omega_e)_z \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

- Or: ${}^0\omega_e = J_\omega \dot{\theta}$.

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

- Or: $\dot{x} = J_v \dot{\theta}$.

- J_ω and J_v are the rotational and linear Jacobian respectively.

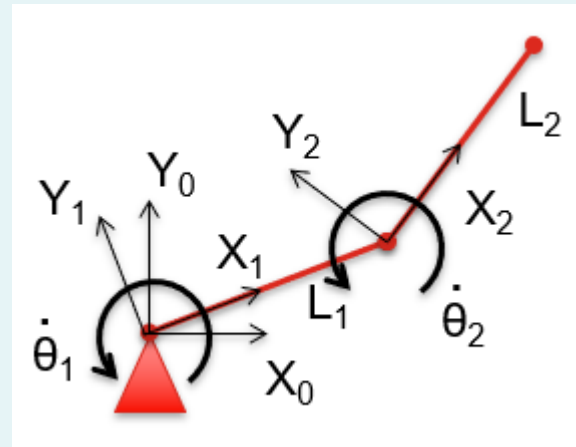
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Another Way to Calculate Velocity (1)

- From simple geometry (or from forward kinematics), we know that the position of the robot tip is:

$${}^0P = \begin{bmatrix} L_1 c_1 + L_2 c_{12} \\ L_1 s_1 + L_2 s_{12} \\ 0 \end{bmatrix}$$



Another Way to Calculate Velocity (2)

- Differentiate the position vector wrt time, using chain rule:

$$\begin{aligned}\frac{d}{dt}({}^0P) &= \begin{bmatrix} \frac{d}{dt}(L_1c_1 + L_2c_{12}) \\ \frac{d}{dt}(L_1s_1 + L_2s_{12}) \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial}{\partial \theta_1}(L_1c_1 + L_2c_{12}) \cdot \frac{\partial \theta_1}{\partial t} + \frac{\partial}{\partial \theta_2}(L_1c_1 + L_2c_{12}) \cdot \frac{\partial \theta_2}{\partial t} \\ \frac{\partial}{\partial \theta_1}(L_1s_1 + L_2s_{12}) \cdot \frac{\partial \theta_1}{\partial t} + \frac{\partial}{\partial \theta_2}(L_1s_1 + L_2s_{12}) \cdot \frac{\partial \theta_2}{\partial t} \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} (-L_1s_1 - L_2s_{12})\dot{\theta}_1 - L_2s_{12}\dot{\theta}_2 \\ (L_1c_1 + L_2c_{12})\dot{\theta}_1 + L_2c_{12}\dot{\theta}_2 \\ 0 \end{bmatrix} = \begin{bmatrix} -L_1s_1\dot{\theta}_1 - L_2s_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ L_1c_1\dot{\theta}_1 + L_2c_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}\end{aligned}$$

Jacobian (1)

- This result is not surprising, as velocity is the derivatives of position.
- We can perform this derivation in a systematic way.
- First we introduce a **generalized joint coordinate**.

$$q_i = \begin{cases} \theta_i & \text{if revolute} \\ d_i & \text{if prismatic} \end{cases}$$

Jacobian (2)

- Using this notation, the joint coordinate vector is:

$$q = [q_1, q_2, \dots, q_n]$$

- The Cartesian position of the tip of the arm is therefore:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} f_x(q) \\ f_y(q) \\ f_z(q) \end{bmatrix}$$

Jacobian (3)

- Differentiation gives the **Cartesian velocity** of the tip of the arm:

$$\begin{aligned}\dot{x} &= \frac{\partial f_x}{\partial q_1} \cdot \dot{q}_1 + \cdots + \frac{\partial f_x}{\partial q_n} \cdot \dot{q}_n \\ \dot{y} &= \frac{\partial f_y}{\partial q_1} \cdot \dot{q}_1 + \cdots + \frac{\partial f_y}{\partial q_n} \cdot \dot{q}_n \\ \dot{z} &= \frac{\partial f_z}{\partial q_1} \cdot \dot{q}_1 + \cdots + \frac{\partial f_z}{\partial q_n} \cdot \dot{q}_n\end{aligned}$$

Jacobian (4)

- The partial differentiations can be written in a matrix form:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_x}{\partial q_1} & \frac{\partial f_x}{\partial q_2} & \dots & \frac{\partial f_x}{\partial q_n} \\ \frac{\partial f_y}{\partial q_1} & \frac{\partial f_y}{\partial q_2} & \dots & \frac{\partial f_y}{\partial q_n} \\ \frac{\partial f_z}{\partial q_1} & \frac{\partial f_z}{\partial q_2} & \dots & \frac{\partial f_z}{\partial q_n} \end{bmatrix} \cdot \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

- Or:

$$\dot{x} = J_v(q)\dot{q}$$

- Where J_v is the linear Jacobian matrix.

Jacobian (5)

- Note: This method is suitable for **linear velocity**, but not for rotational velocity.
- Also, remember that the **Jacobian is linear but time varying**.
 - This means, the joint rates are related to the velocity of the tip in a linear manner, but this relationship is only **instantaneous**.
 - At the next instant, we have a new Jacobian and new relationship.

Jacobian (6)

- Looking back at our 2-link robot example:

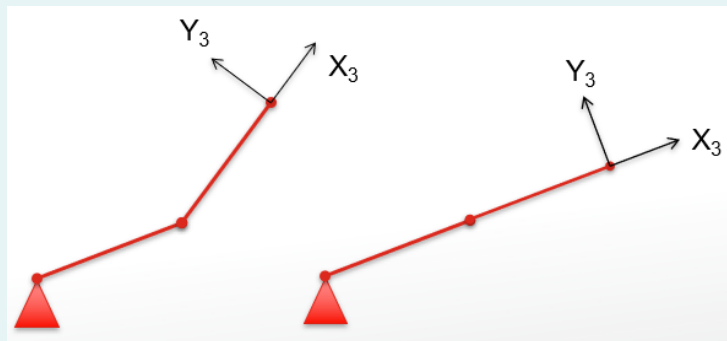
$${}^0P = \begin{bmatrix} L_1 c_1 + L_2 c_{12} \\ L_1 s_1 + L_2 s_{12} \\ 0 \end{bmatrix}$$
$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial q_1} (L_1 c_1 + L_2 c_{12}) & \frac{\partial}{\partial q_2} (L_1 c_1 + L_2 c_{12}) \\ \frac{\partial}{\partial q_1} (L_1 s_1 + L_2 s_{12}) & \frac{\partial}{\partial q_2} (L_1 s_1 + L_2 s_{12}) \\ \frac{\partial}{\partial q_1} (0) & \frac{\partial}{\partial q_2} (0) \end{bmatrix} \cdot \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}$$
$$= \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}$$

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Kinematic Singularities (1)

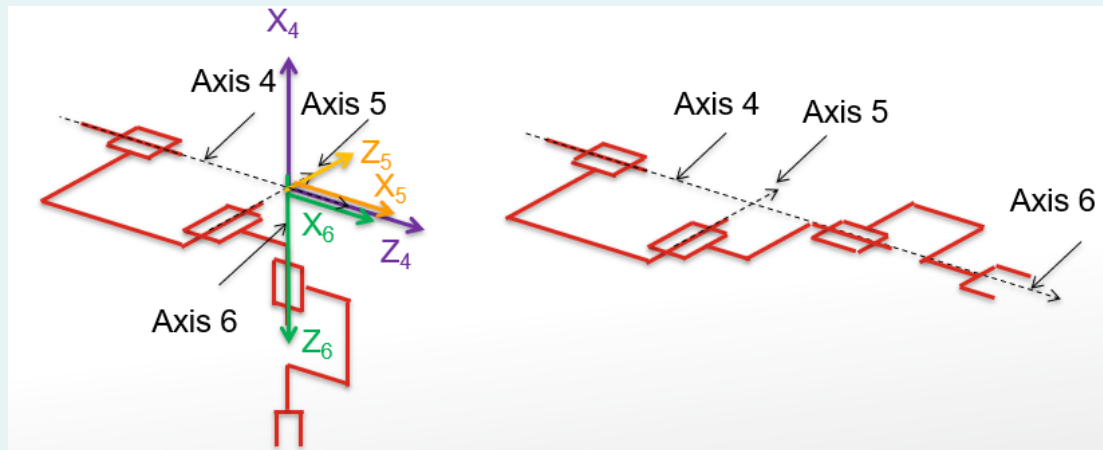
- Kinematic **singularity** happens when the end-effector **loses the ability** to move in a direction, or to rotate about a direction.
- E.g. Two-link robot:



- In the left figure, the end-effector can move in any direction instantly.
- In the right figure, the end-effector loses the ability to move in the x -direction.

Kinematic Singularities (2)

- Another example: The last 3 joints (the wrist) of a robot:



- In the left figure, the end-effector can rotate about a “vertical axis”.
- In the right figure, when axes 4 and 6 are collinear, the end-effector loses ability to rotate about the axis.

Kinematic Singularities (3)

- Mathematically, these singularities happen when the **Jacobian matrix becomes non-invertible / singular**.
- We knew: $v = J(q)\dot{q}$.
- The inverse question is: If we require a certain movement v , what should the joint rate $\dot{q}$ be?
- This can be obtained via: $\dot{q} = J^{-1}(q)v$
- If the **Jacobian matrix is not invertible** or ill-conditioned (determinant close to zero), then we need an **infinitely big joint rate** to obtain v .
- Not possible or practical.

Kinematic Singularities (4)

- For the two-link robot example, the Jacobian was:

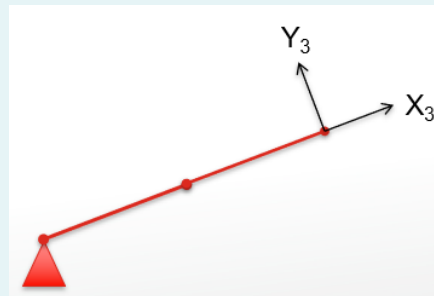
$$\begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 0 & 0 \end{bmatrix}$$

- Its determinant (ignoring the last row with zeros) is:

$$-L_1 L_2 s_1 c_{12} - L_2^2 s_{12} c_{12} + L_1 L_2 c_1 s_{12} + L_2^2 s_{12} c_{12} = L_1 L_2 s_2$$

- The determinant is zero (the Jacobian becomes singular) when $s_2 = 0$ or

$$\theta_2 = k\pi.$$

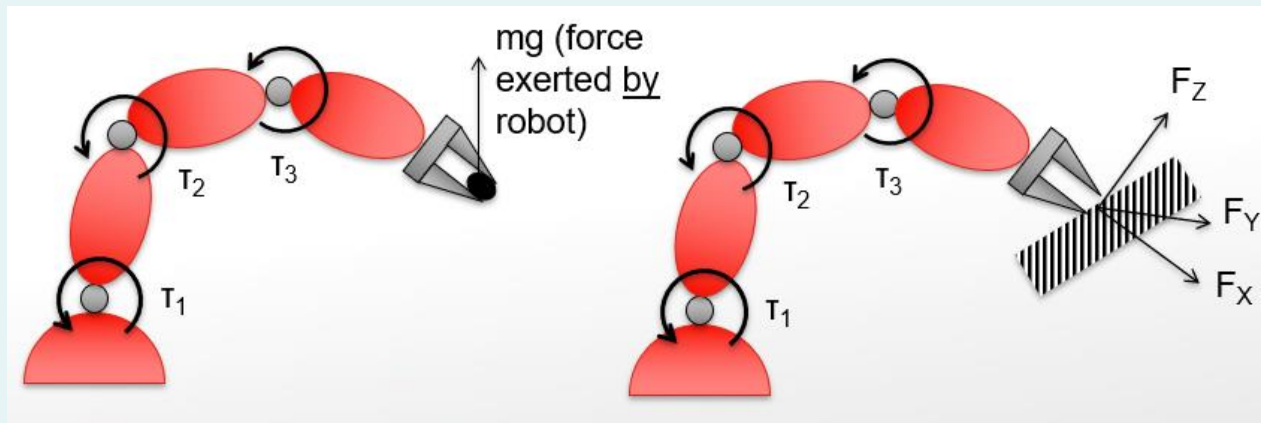


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Static Forces in Manipulators (1)

- The question we would like to answer is as follows:
 - The robot is **holding an object with force mg** (left), or
 - The robot is **pushing the environment with force F** (right).
 - What would be the **joint torques** needed to keep the system in **static equilibrium**?



Static Forces in Manipulators (2)

- Good news! The answer uses the **same Jacobian** from the velocity question:

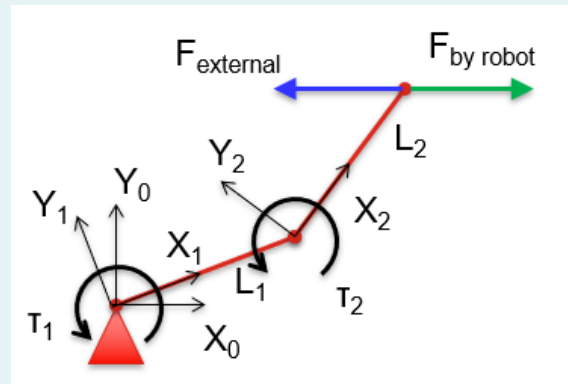
$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \vdots \\ \tau_n \end{bmatrix} = {}^0J_v^T \cdot \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}$$

- Or in short:

$$\tau = {}^0J_v^T \cdot F$$

Static Forces in Manipulators (3)

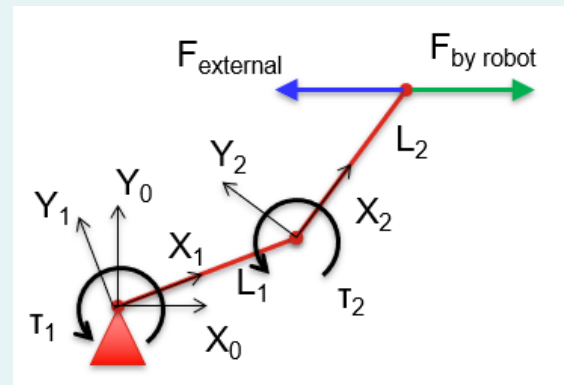
- Example: The same 2-link robot as earlier. A force F_{external} acts on the tip of the robot horizontally wrt $\{0\}$ towards the left.
- To maintain **static equilibrium**, the robot exerts a force with the same magnitude but in the opposite direction (towards the right).
- What **torques** are provided by the motors to produce the robot force?



Static Forces in Manipulators (4)

- Answer:

$$\begin{aligned}
 \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} &= {}^0J_v^T \cdot \begin{bmatrix} {}^0F_x \\ {}^0F_y \end{bmatrix} \\
 &= \begin{bmatrix} -L_1s_1 - L_2s_{12} & L_1c_1 + L_2c_{12} \\ -L_2s_{12} & L_2c_{12} \end{bmatrix} \cdot \begin{bmatrix} F_{\text{by robot}} \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} (-L_1s_1 - L_2s_{12})F_{\text{by robot}} \\ (-L_2s_{12})F_{\text{by robot}} \end{bmatrix}
 \end{aligned}$$

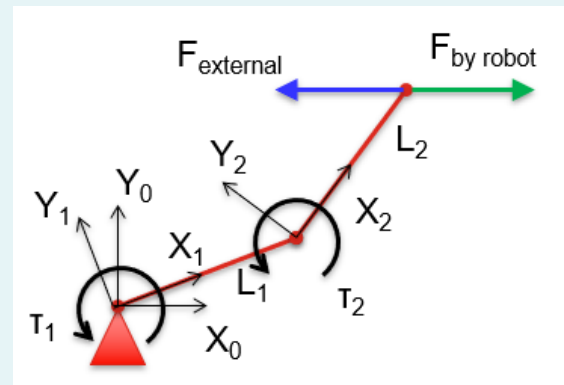


Static Forces in Manipulators (5)

- Numerical: If $\theta_1 = 30^\circ, \theta_2 = 30^\circ, F_{\text{by robot}} = 10\text{N}, L_1 = L_2 = 1\text{m}$:

$$\begin{aligned} \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} &= \begin{bmatrix} (-L_1 s_1 - L_2 s_{12}) F_{\text{by robot}} \\ (-L_2 s_{12}) F_{\text{by robot}} \end{bmatrix} \\ &= \begin{bmatrix} (-0.5\text{m} - 0.866\text{m}) 10\text{N} \\ (-0.866\text{m}) 10\text{N} \end{bmatrix} = \begin{bmatrix} -13.66\text{Nm} \\ -8.66\text{Nm} \end{bmatrix} \end{aligned}$$

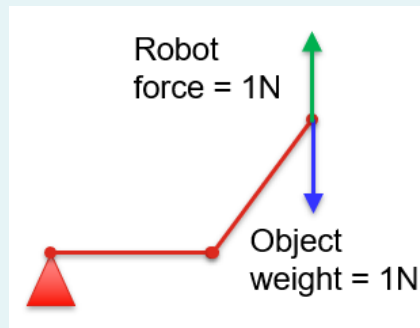
- The negative signs in the answer make sense: The **motor torques are in clockwise** direction so that robot force points to the right.



Singularity and Static Forces (1)

- Still the same robot but different configuration and forces.
- Case 1: $\theta_1 = 0^\circ, \theta_2 = 60^\circ, F_{\text{by robot}} = 1\text{N}$.

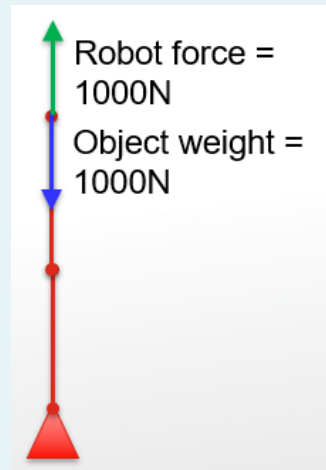
$$\begin{aligned} \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} &= \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & L_1 c_1 + L_2 c_{12} \\ -L_2 s_{12} & L_2 c_{12} \end{bmatrix} \cdot \begin{bmatrix} {}^0F_x \\ {}^0F_y \end{bmatrix} \\ &= \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & L_1 c_1 + L_2 c_{12} \\ -L_2 s_{12} & L_2 c_{12} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1\text{N} \end{bmatrix} \\ &= \begin{bmatrix} (L_1 c_1 + L_2 c_{12})1\text{N} \\ (L_2 c_{12})1\text{N} \end{bmatrix} \\ &= \begin{bmatrix} (1\text{m} + 0.5\text{m})1\text{N} \\ (0.5\text{m})1\text{N} \end{bmatrix} = \begin{bmatrix} 1.5\text{Nm} \\ 0.5\text{Nm} \end{bmatrix} \end{aligned}$$



Singularity and Static Forces (2)

- Case 2: $\theta_1 = 90^\circ, \theta_2 = 0^\circ, F_{\text{by robot}} = 1000\text{N}$.

$$\begin{aligned} \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} &= \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & L_1 c_1 + L_2 c_{12} \\ -L_2 s_{12} & L_2 c_{12} \end{bmatrix} \cdot \begin{bmatrix} {}^0F_x \\ {}^0F_y \end{bmatrix} \\ &= \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & L_1 c_1 + L_2 c_{12} \\ -L_2 s_{12} & L_2 c_{12} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1000\text{N} \end{bmatrix} \\ &= \begin{bmatrix} (L_1 c_1 + L_2 c_{12})1000\text{N} \\ (L_2 c_{12})1000\text{N} \end{bmatrix} \\ &= \begin{bmatrix} (0\text{m} + 0\text{m})1\text{N} \\ (0\text{m})1\text{N} \end{bmatrix} = \begin{bmatrix} 0\text{Nm} \\ 0\text{Nm} \end{bmatrix} \end{aligned}$$



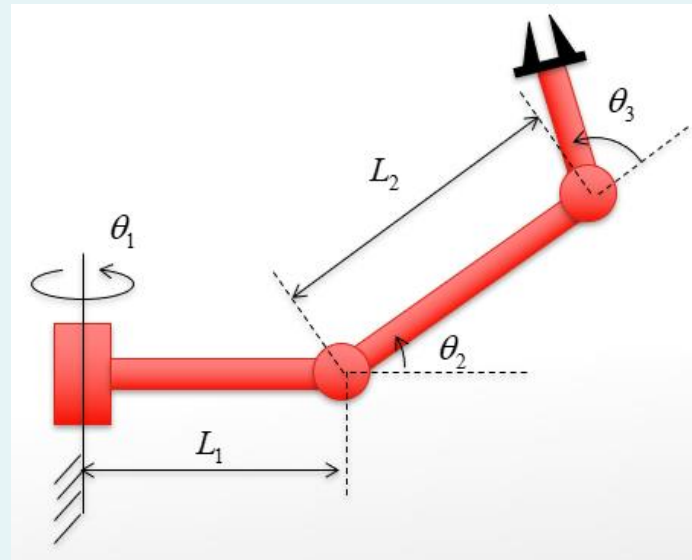
- At **singularity**, no torque is needed to withstand large forces → **mechanical advantage**!

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Question 1

- Find the Jacobian of the manipulator shown on the right. (You should already have some information about this robot in earlier tutorials).
- Write it in terms of frame {3} at the wrist of robot.
 - Velocity propagation method.
 - Differentiation of kinematic equations.
- Write also in terms of frame {4} at the tip of hand, having same orientation as {3}.



Question 2

- A 2-link manipulator has the following Jacobian:

$${}^0J_v = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix}$$

- If we ignore gravity, what are the joint torques required so that the manipulator can apply a static force of

$${}^0F = 10\hat{X}_0$$

Thank you for your attention!

Any questions?